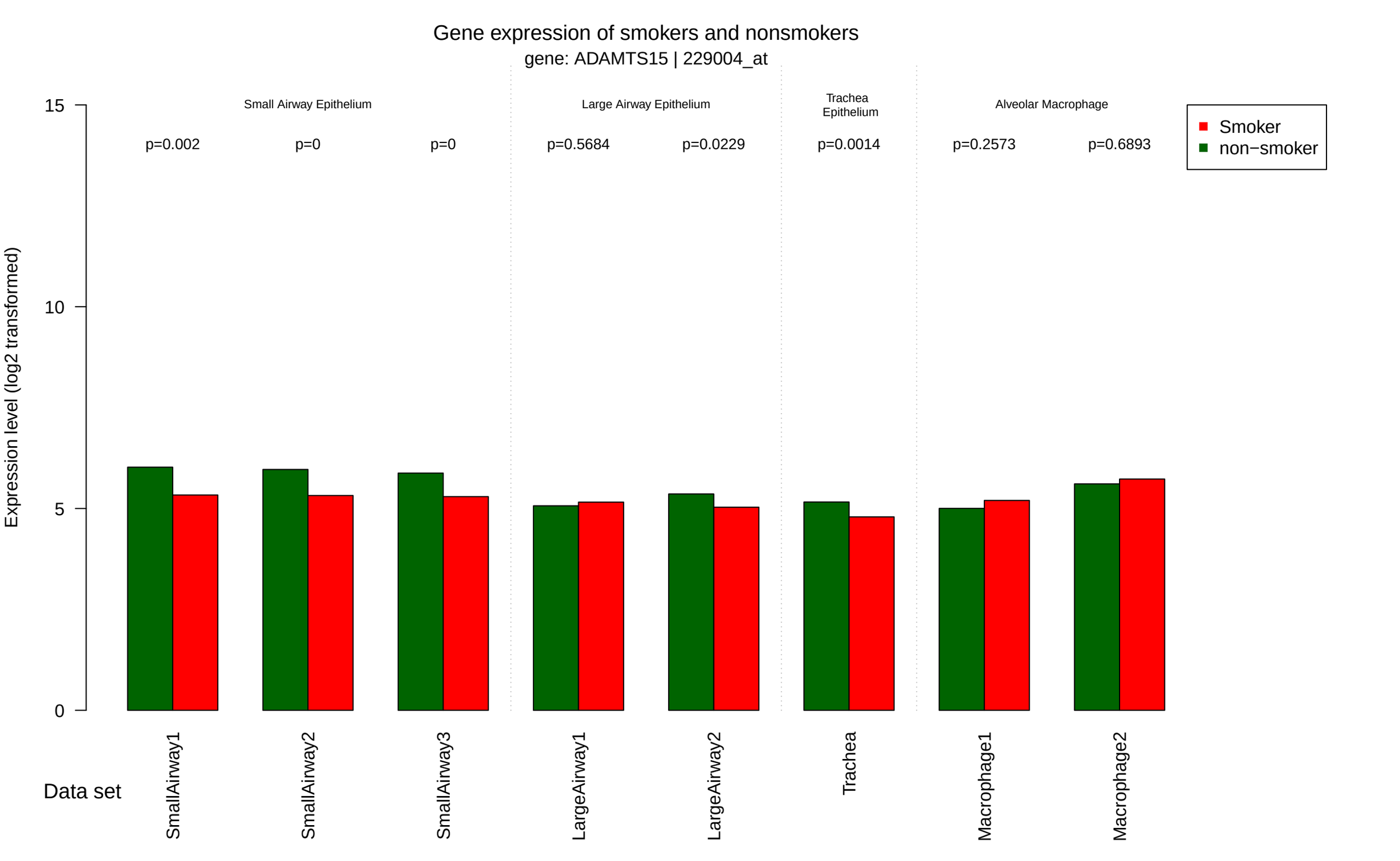
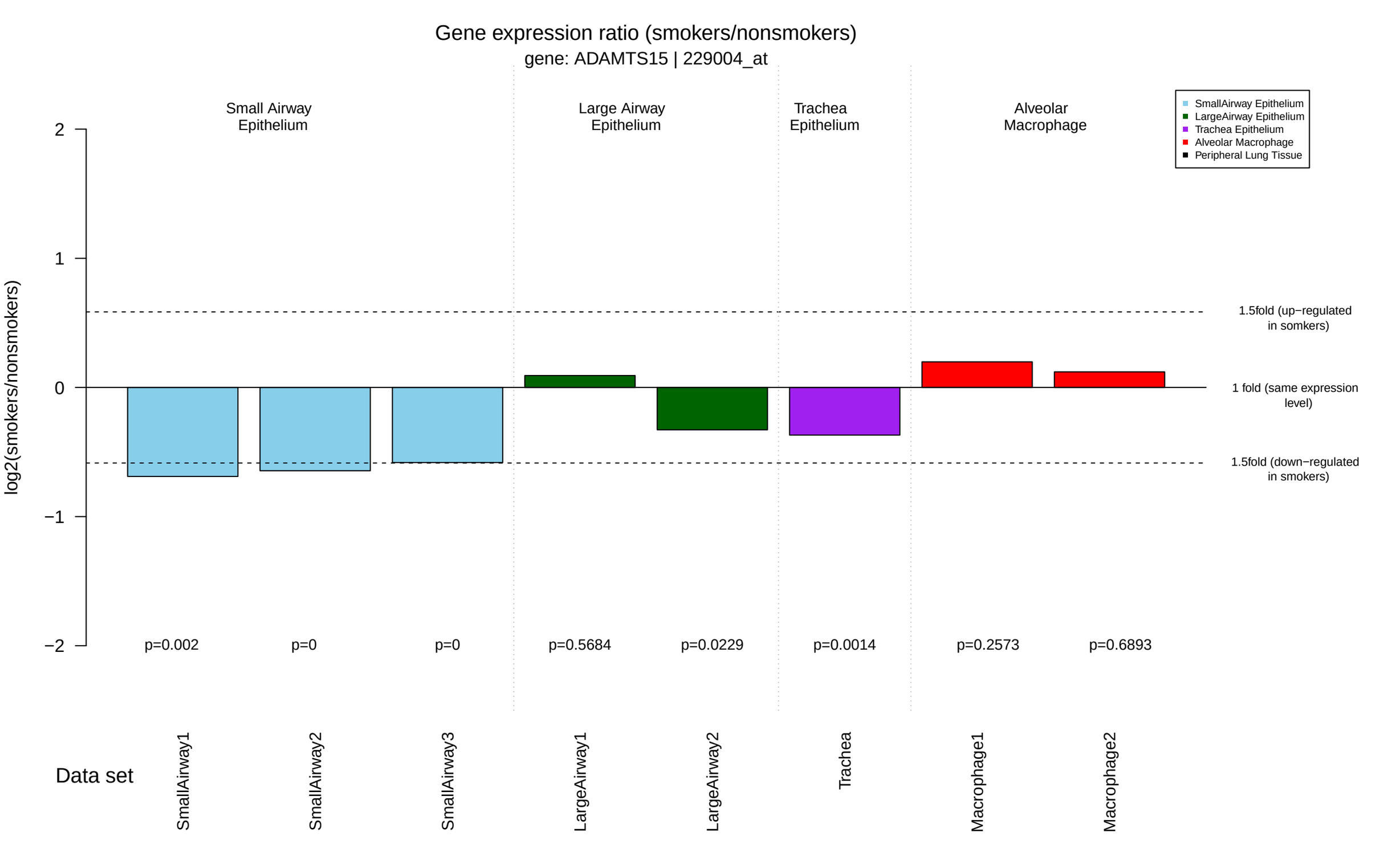
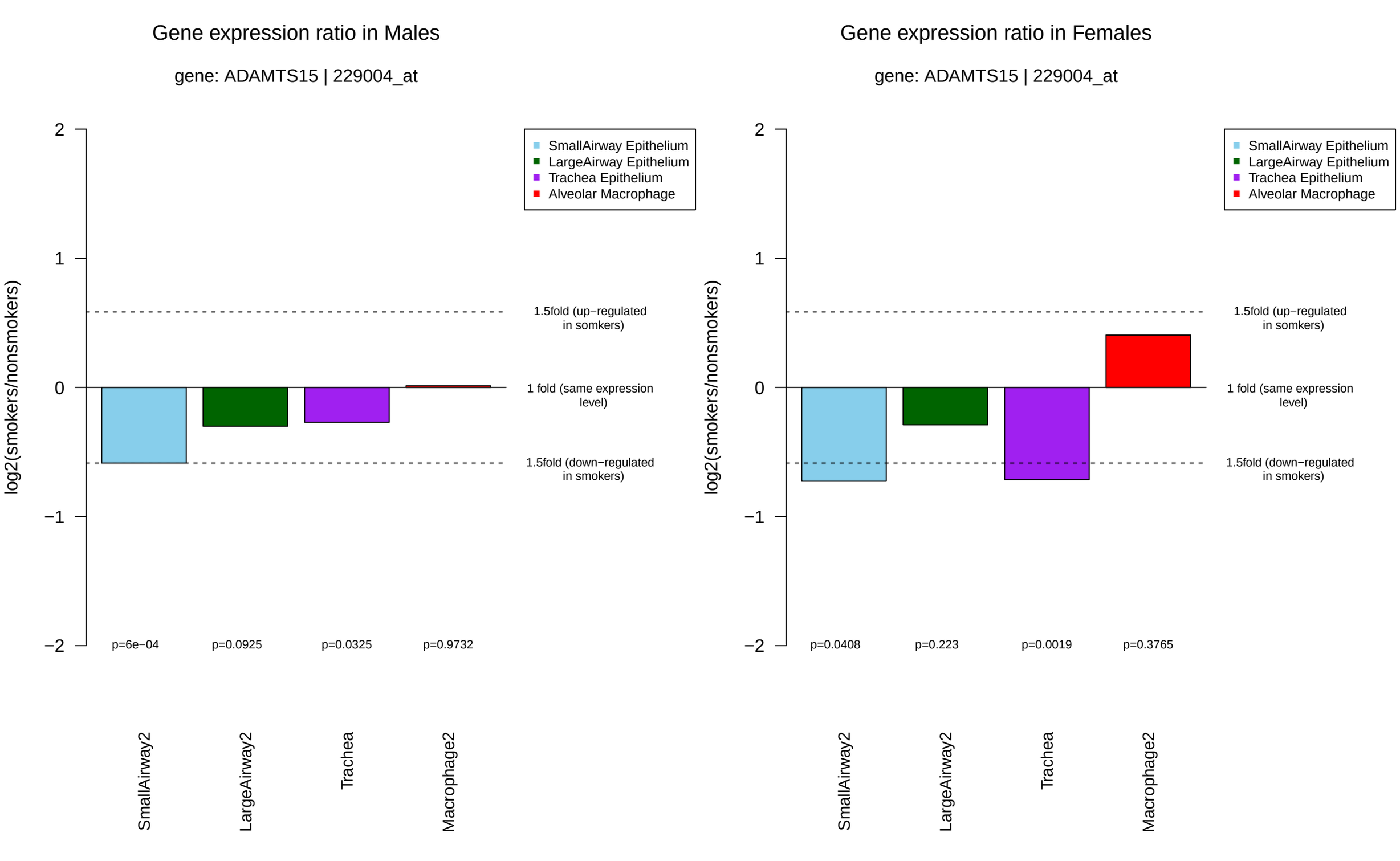
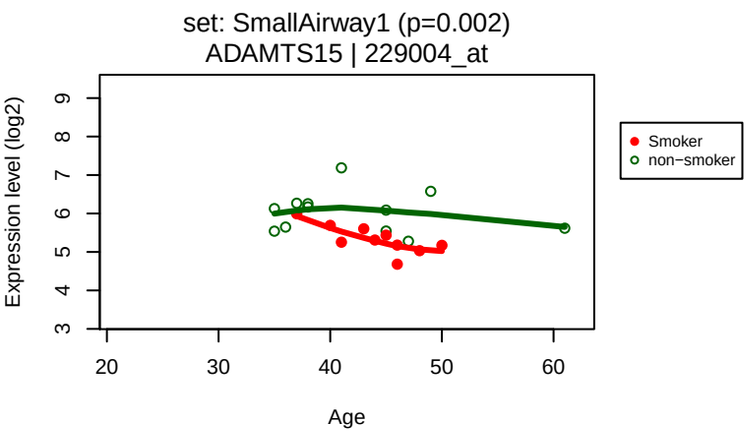
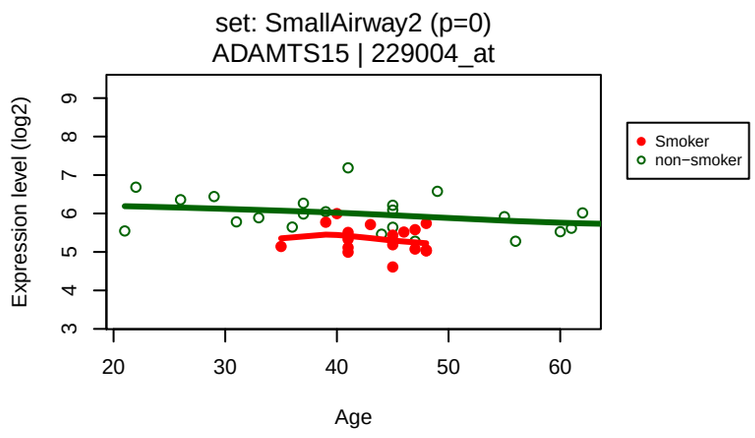
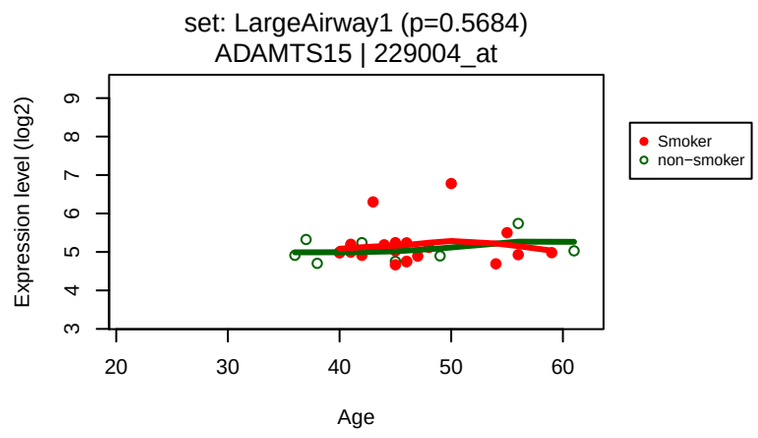

Gene expression of smokers and nonsmokers
gene: ADAMTS15 | 229004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.002
p=0
p=0
p=0.5684
p=0.0229
p=0.0014
p=0.2573
p=0.6893
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

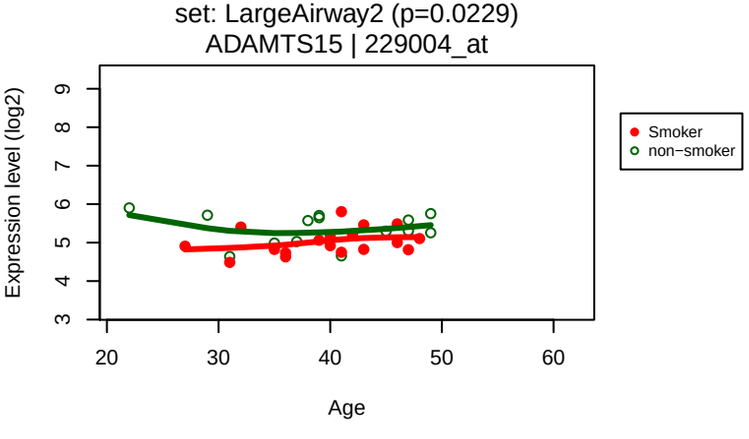

Gene expression ratio (smokers/nonsmokers)
gene: ADAMTS15 | 229004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0
p=0
p=0.5684
p=0.0229
p=0.0014
p=0.2573
p=0.6893
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

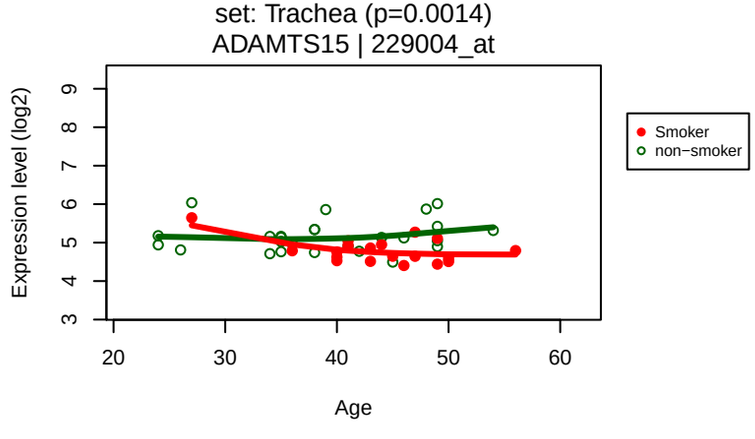

Gene expression ratio in Males
gene: ADAMTS15 | 229004_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.0925
p=0.0325
p=0.9732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADAMTS15 | 229004_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0408
p=0.223
p=0.0019
p=0.3765
SmallAirway2
LargeAirway2
Trachea
Macrophage2

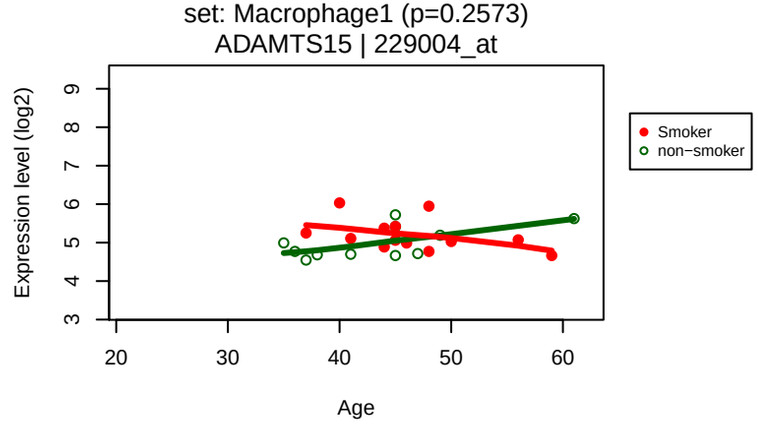

set: SmallAirway1 (p=0.002)
ADAMTS15 | 229004_at
Expression level (log2)
Age
Smoker
non−smoker

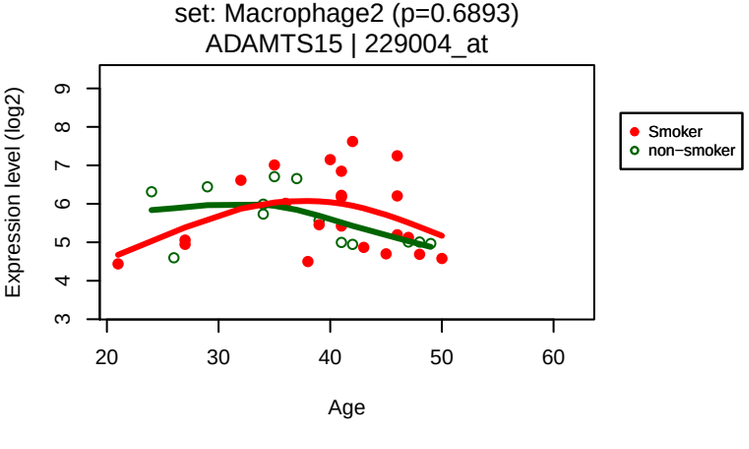

set: SmallAirway2 (p=0)
ADAMTS15 | 229004_at
Expression level (log2)
Age
Smoker
non−smoker

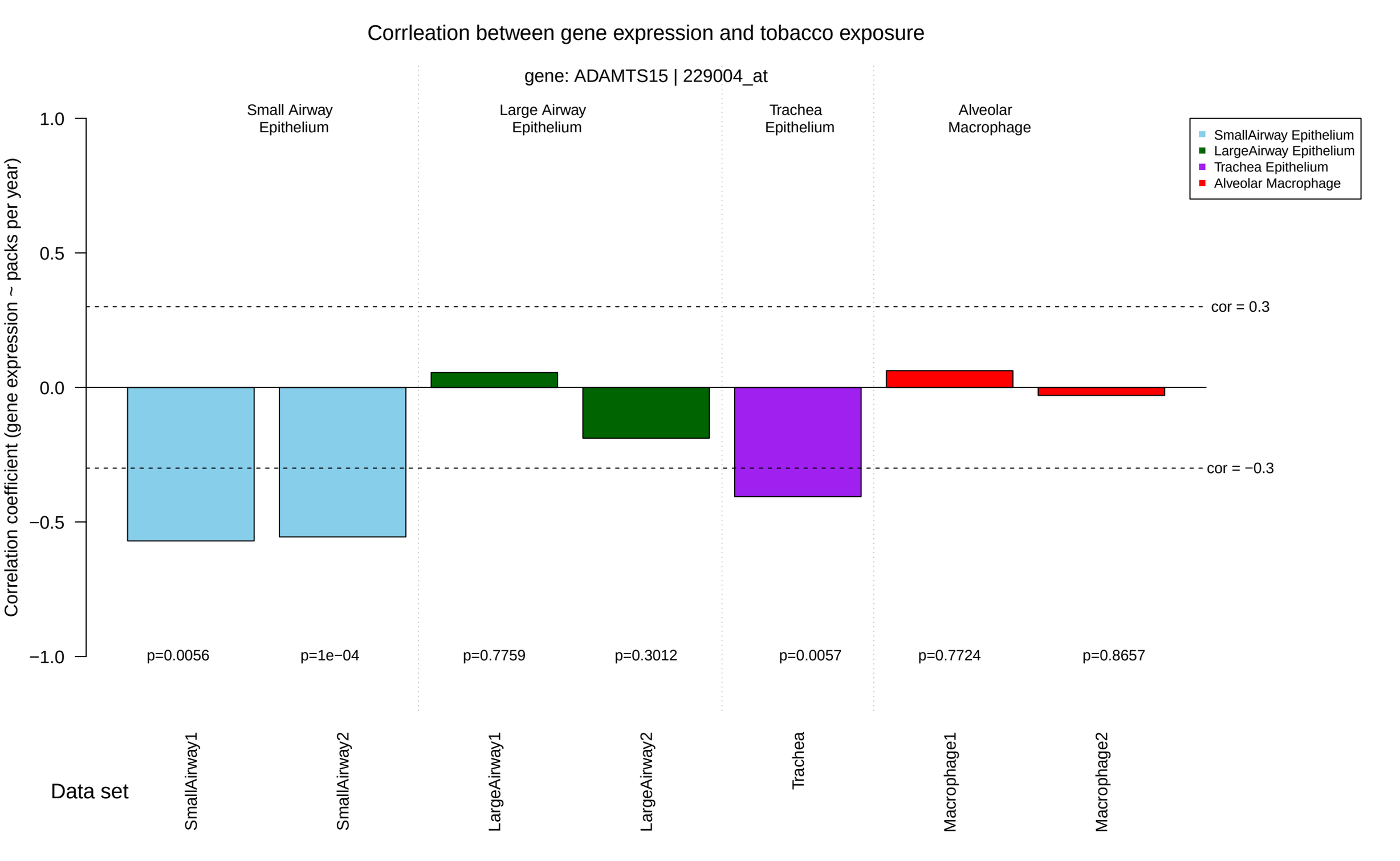

set: LargeAirway1 (p=0.5684)
ADAMTS15 | 229004_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0229)
ADAMTS15 | 229004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0014)
ADAMTS15 | 229004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2573)
ADAMTS15 | 229004_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6893)
ADAMTS15 | 229004_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADAMTS15 | 229004_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0056
p=1e−04
p=0.7759
p=0.3012
p=0.0057
p=0.7724
p=0.8657
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2